Nazorat testi · *Aralash test · 4-variant* — *Milliy sertifikat 2026. Kimyo*

Vaqt: *180 daqiqa*

Davriy jadvaldan foydalanish mumkin.

1. A moddaning parchalanish reaksiyasida konsentratsiyaning vaqt bo‘yicha o‘zgarishi jadvalda berilgan:

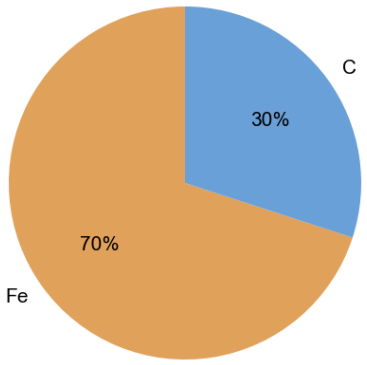
t, min	c, mol/l
0	1,6
2	1,44
4	1,28
6	1,12

2-minutdan 6-minutgacha o‘rtacha tezlikni mol/(l·min) **hisoblang.**

A) *0,16* B) *0,08* C) *0,04* D) *0,32*

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2. Qotishmaning massa bo‘yicha tarkibi doira diagrammasida ko‘rsatilgan.



Qotishmadagi Fe va C atomlari sonining nisbati (Fe uchun C ga **bo‘lingan)ni toping.**

A) *0,5* B) *2,33* C) *4,67* D) *0,43*

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3. Massa soni 226, zaryadi 88 bo‘lgan yadro massa soni 206, zaryadi 82 bo‘lgan yadroga aylandi (α va β^- parchalanish). Nechta α-zarra chiqarilgan?

A) *6* B) *4* C) *7* D) *5*

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4. 30 l CO va 22,5 l O_2 aralashmasi reaksiyaga kirishdi ($2CO + O_2 \rightarrow 2CO_2$, hamma moddalar bir xil sharoitda gaz holida). Reaksiyadan keyingi gaz aralashmasining **hajmini (l) aniqlang.**

A) *52,5* B) *30* C) *37,5* D) *7,5*

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5. $KMnO_4$ tarkibida Mn ning oksidlanish **darajasini aniqlang.**

A) -2 B) $+7$ C) $+4$ D) $+3$ *va* $+5$

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6. X^{2+} ionida 18 ta elektron va 23 ta neytron bor. X atomining nisbiy atom **massasini aniqlang.**

A) *43* B) *45* C) *41* D) *20*

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7. $2NO + O_2 \rightleftarrows 2NO_2$ reaksiyada muvozanat qaror topganda $[NO] = 1$, $[O_2] = 1$, $[NO_2] = 4$ mol/l bo‘ldi. O_2 ning boshlang‘ich **konsentratsiyasini (mol/l) aniqlang.**

A) *3* B) *1* C) *5* D) *6*

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8. $2SO_2 + O_2 \rightleftarrows 2SO_3 + Q$ muvozanatni o‘ngga siljitadigan **o‘zgarishlarni toping**: 1) katalizator qo‘shish; 2) haroratni pasaytirish; 3) bosimni oshirish; 4) bosimni pasaytirish.

A) *1, 2* B) *3, 4* C) *2, 3* D) *1, 3, 4*

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9. $CuSO_4 \cdot 5H_2O$ kristallogidratida kristallizatsion suvning massa **ulushini (%) aniqlang.**

A) *36* B) *64* C) *18* D) *56,2*

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10. $AgNO_3$ eritmasi inert elektrodlar yordamida 2 A tok kuchi bilan 965 s elektroliz qilindi. Anodda ajralgan O_2 hajmi (l, n.sh.) **ni toping.**

A) *0,224* B) *0,056* C) *0,002* D) *0,112*

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11. Zarracha haqida ma’lumot jadvalda berilgan:

Zarracha tarkibi	Soni
proton	30
neytron	34
elektron	29

Shu zarrachaning massa **sonini aniqlang.**

A) *64* B) *65* C) *59* D) *60*

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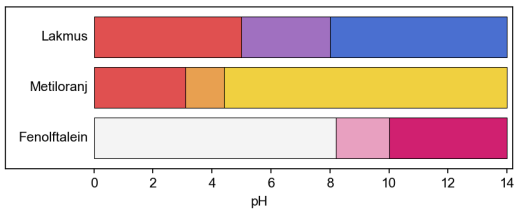
12. Qaysi juftliklarda reaksiya boradi? 1) Mg va $FeSO_4$; 2) Cu va $AgNO_3$; 3) Zn va $CuSO_4$; 4) Cu va $ZnSO_4$

A) *2, 3* B) *1, 2, 3* C) *1, 2* D) *2, 4*

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13. Rasmda indikatorlarning pH ga bog‘liq rangi ko‘rsatilgan.

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pH = 2 bo‘lgan eritmada lakmus qanday rangga kiradi?
A) *zarg‘aldoq* B) *binafsha*
C) *qizil* D) *och pushti*

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14. Qaysi juftliklar o‘zaro reaksiyaga kirishadi? 1) $CuSO_4$ va NaOH; 2) Zn va NaOH; 3) $MgCl_2$ va KNO_3; 4) Cu va HNO_3(kons.)
A) *1, 4* B) *1, 3* C) *2, 3, 4* D) *1, 2, 4*

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15. $2KClO_3 \rightarrow 2KCl + 3O_2$ reaksiyasida 245 g $KClO_3$ ishtirok etdi. O_2 **hajmini (l, n.sh.) toping.**
A) *29,87* B) *67,2* C) *134,4* D) *33,6*

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16. K_2SO_4 tarkibidagi kaliy ning massa **ulushini (%) aniqlang.**
A) *44,8* B) *22,4* C) *89,7* D) *2,2*

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17. $AgNO_3$ qizdirilganda qanday moddalar hosil bo‘ladi?
A) *CuO, NO_2 va O_2* B) *KNO_2 va O_2*
C) *Ag, NO_2 va O_2* D) *ZnO, NO_2 va O_2*

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18. Reaksiya tenglamasi: $FeS + 2HCl \rightarrow FeCl_2 + H_2S$. Reaksiyaga 11,2 l (n.sh.) HCl kirishdi. $FeCl_2$ **massasini (g) hisoblang.**
A) *127* B) *63,5* C) *31,75* D) *15,88*

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19. Reaksiya tenglamasi: $NH_4Cl + NaOH \rightarrow NaCl + NH_3 + H_2O$. Reaksiyaga tarkibida 60% aralashma bo‘lgan 40 g namuna (asosiy modda NaOH) kirishdi. NH_3 **hajmini (l, n.sh.) hisoblang.**
A) *17,92* B) *8,96* C) *4,48* D) *13,44*

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20. C_3H_7OH tarkibli spirtlarning izomerlari soni nechta?
A) *3* B) *1* C) *4* D) *2*

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21. Alken, alkadiyen va alkinlar haqida qaysi fikrlar to‘g‘ri? 1) Kauchuk izoprenning polimeri; 2) Alkenlarda π-bog‘ yo‘q; 3) Alkinlar uchun umumiy formula C_nH_{2n+2}; 4) Propenga HBr birikishi Markovnikov qoidasiga bo‘ysunadi
A) *1, 4* B) *2, 3, 4* C) *1, 2, 3* D) *1, 3*

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22. $C_2H_2 + H_2O \rightarrow CH_3CHO$ reaksiyasida 52 g C_2H_2 ishtirok etdi. CH_3CHO **massasini (g) toping.**
A) *176* B) *44* C) *132* D) *88*

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23. Spirt va fenollar haqida qaysi fikrlar to‘g‘ri? 1) Fenol $FeCl_3$ bilan binafsha rang beradi; 2) Uchlamchi spirtlar oson oksidlanadi; 3) Fenol $Cu(OH)_2$ ni eritib ko‘k kompleks hosil qiladi; 4) Glitserin $Cu(OH)_2$ bilan ko‘k kompleks hosil qiladi
A) *1, 4* B) *1, 2, 3* C) *1, 3, 4* D) *1, 3*

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24. 18 g glyukoza Ag_2O ning ammiakli eritmasi bilan to‘liq oksidlanganda hosil bo‘lgan kumush **massasini (g) aniqlang.**
A) *10,8* B) *43,2* C) *21,6* D) *18*

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25. Karbon kislotalar haqida qaysi fikrlar to‘g‘ri? 1) Chumoli kislota "kumush ko‘zgu" reaksiyasini beradi; 2) Sirka angidrid ikki molekula sirka kislotadan suv chiqib hosil bo‘ladi; 3) Benzoy kislota Na_2CO_3 bilan reaksiyaga kirishmaydi; 4) Qattiq suvda sovun yaxshi ko‘piklanadi
A) *3, 4* B) *2, 3* C) *1, 2, 3* D) *1, 2*

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26. $CH_3COOCH_3 + NaOH \rightarrow CH_3COONa + CH_3OH$ jarayoni uchun 20 g NaOH olingan (unum 90%). CH_3COONa **massasini (g) aniqlang.**
A) *41* B) *36,9* C) *73,8* D) *18,45*

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27. Quyidagi birikmalardan qaysi biri karbon kislota sinfiga kiradi?
A) *C_2H_5OH* B) *$C_{12}H_{22}O_{11}$* C) *CH_3COOH* D) *$C_6H_{12}O_6$*

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28. 11,7 g valin HCl bilan reaksiyaga kirishganda hosil bo‘lgan tuz **massasini (g) aniqlang.**
A) *11,7* B) *3,65* C) *30,7* D) *15,35*

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29. $C_2H_2 + HCl \rightarrow C_2H_3Cl$ reaksiyasida 44,8 l (n.sh.) HCl ishtirok etdi. C_2H_2 **hajmini (l, n.sh.) toping.**
A) *89,6* B) *22,4* C) *67,2* D) *44,8*

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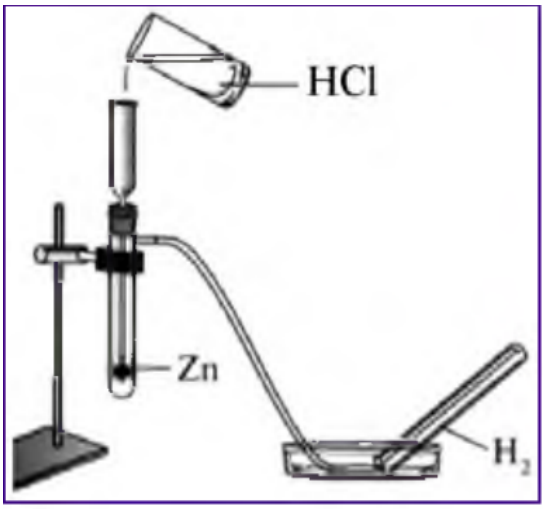
30. Cu^{2+} ionini aniqlash uchun qaysi reagent mos va qanday belgi kuzatiladi? A) *NaOH: oq-yashil cho‘kma ($Fe(OH)_2$)*
B) *NaOH: havorang cho‘kma ($Cu(OH)_2$)*
C) *HCl: oq cho‘kma (AgCl)*
D) *$AgNO_3$: och sariq cho‘kma (AgBr)*

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31. 250 g 10% li $CuSO_4$ eritmasidan 50 g suv bug‘latildi. Qolgan eritmadagi tuzning massa **ulushini (%) toping.**
A) *8,33* B) *12,5* C) *2* D) *10*

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32. Rasmdagi asbobda 6,5 g rux xlorid kislota bilan reaksiyaga kirishdi ($Zn + 2HCl \rightarrow ZnCl_2 + H_2$). Ajralgan vodorodning **hajmini (l, n.sh.) aniqlang.**



A) *4,48* B) *1,12* C) *2,24* D) *0,1* —

Topshiriqlar (33–35) va javob variant (A–F)larini o‘zaro to‘g‘ri moslashtiring. NaBr va KCl dan iborat 12,6 g aralashma suvda eritilib, mo‘l $AgNO_3$ eritmasi qo‘shildi; hosil bo‘lgan cho‘kmalarning umumiy massasi 23,75 g.	A *5,15* B *7,73* C *14,35* D *11,48* E *255* F *127,5*
33. Aralashmadagi NaBr massasini (g) aniqlang.	
34. Sarflangan 10% li $AgNO_3$ eritmasining massasini (g) aniqlang (M($AgNO_3$) = 170 g/mol).	
35. Eritmada hosil bo‘lgan nitratlarning umumiy massasini (g) aniqlang.	

36. $H_2S + HNO_3 \rightarrow H_2SO_4 + NO_2 + H_2O$ reaksiya tenglamasini tenglang. Reagentlar (chap qism) koeffitsiyentlari **yig‘indisini toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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37. 1 l hajmli yopiq idishda 4 mol N_2O_4 qizdirilganda $N_2O_4 \rightleftarrows 2NO_2$ muvozanati qaror topdi. Muvozanat aralashmasining vodorodga nisbatan zichligi 36,8 ga teng. Shu haroratdagi muvozanat **konstantasini (K, mol/l) hisoblang.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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38. Alyuminiy va kalsiy dan iborat 6,7 g aralashma mo‘l xlorid kislotada eritilganda 5,6 l (n.sh.) vodorod ajraldi. Aralashmadagi alyuminiyning massa **ulushini (%) aniqlang.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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39. Etilen va asetilendan iborat 4,48 l (n.sh.) aralashma bromli suvdan o‘tkazilganda 40 g brom yutildi. Aralashmadagi asetilenning **hajmini (l, n.sh.) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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40. 17,8 g tristearin mo‘l NaOH eritmasi bilan to‘liq gidrolizlandi (sovun unumi 90%). Hosil bo‘lgan natriy stearat (M = 306 g/mol) **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

YOZMA ISH

41. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

14,3 g $Na_2CO_3 \cdot 10H_2O$ 400 g suvda eritildi.
1)Hosil bo‘lgan eritmadagi Na_2CO_3 ning massa ulushini (%) toping.
2)Eritmaga mo‘l $CaCl_2$ qo‘shildi. Hosil bo‘lgan cho‘kma massasini (g) toping. Eritmadagi kation miqdorini (mol) ham toping.
3)Xuddi shunday eritmaga mo‘l HCl qo‘shilganda ajralgan gazning hajmini (l, n.sh.) hisoblang.
4)Eritmani to‘liq neytrallash ($Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$) uchun 10% li HCl eritmasidan necha gramm kerak? Dastlabki eritma massasini (g) ham ayting.
5)Kristallogidratdagi kristallizatsion suvning massa ulushini (%) va unda suv molekulalari sonini toping. Shu miqdor kristallogidrat suvsizlantirilganda massa necha gramm kamayadi?

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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42. Quyidagi o‘zgarishlar zanjirida X, Y, Z noma’lum organik moddalar:

$$C_6H_{12}O_6 \xrightarrow{\text{bijg‘ish}} X \xrightarrow{H_2SO_4\ (\text{kons.}),\ 170\ ^\circ C} Y \xrightarrow{Br_2} C_2H_4Br_2$$

1)$C_6H_{12}O_6 \rightarrow X$: reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.
2)$X \rightarrow Y$: reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.
3) $Y \rightarrow C_2H_4Br_2$: reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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43. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

5,6 g temir mo‘l xlorda yondirildi. Hosil bo‘lgan tuz suvda eritilib, avval ishqor bilan, so‘ng boshqa porsiyasi KI eritmasi bilan ta’sirlashtirildi.
1)Reaksiya tenglamasini yozing. Sarflangan xlor hajmini (l, n.sh.) va hosil bo‘lgan tuz massasini (g) toping. Hosil bo‘lgan $FeCl_3$ miqdorini (mol) ham toping.
2)Tuz eritmasiga mo‘l NaOH qo‘shilganda hosil bo‘lgan cho‘kma massasini (g) aniqlang.
3)Cho‘kma qizdirilganda hosil bo‘ladigan oksid massasini (g) hisoblang.

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4)Tuzning shunday miqdoriga mo‘l KI qo‘shilganda ajraladigan yod massasini (g) toping ($2FeCl_3 + 2KI \rightarrow 2FeCl_2 + 2KCl + I_2$).
5)Hosil bo‘lgan $FeCl_3$ eritmasiga temir qo‘shilib $FeCl_2$ hosil qilindi ($2FeCl_3 + Fe \rightarrow 3FeCl_2$). Sarflangan temir massasini (g) toping.

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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Nazorat testi · *4-variant* **Javoblar kaliti**

Faqat o‘qituvchi uchun.

1–32-savollar

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
B	*A*	*D*	*C*	*B*	*A*	*A*	*C*	*A*	*D*	*A*	*B*	*C*	*D*	*B*	*A*
17	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**
C	*C*	*B*	*D*	*A*	*D*	*A*	*C*	*D*	*B*	*C*	*D*	*D*	*B*	*B*	*C*

33–35-savollar

No	Javob	Qiymat
33	*A*	5,15
34	*E*	255
35	*C*	14,35

36–40-savollar

No	To‘g‘ri javob
36	9
37	1,333
38	40,3
39	1,12
40	16,52

41–43-topshiriqlar: javoblar

41-topshiriq.
1) 1,28%
2) 5 g; 0,05 mol
3) 1,12 l
4) 36,5 g; 414,3 g
5) 62,9%; 10; 9 g

42-topshiriq.
1) $C_6H_{12}O_6$ + bijg‘ish → C_2H_5OH (etanol)
2) C_2H_5OH + H_2SO_4 (kons.), 170 °C → C_2H_4 (etilen)
3) C_2H_4 + Br_2 → $C_2H_4Br_2$ (1,2-dibrometan)

43-topshiriq.
1) 3,36 l; 16,25 g; 0,1 mol
2) 10,7 g
3) 8 g
4) 12,7 g
5) 2,8 g

Mavzular taqsimoti

No	Mavzu	Tekshiriladigan ko‘nikma	Ball
1	Tirkashev VI bob (tezlik, jadval)	Jadval tahlili	1
2	8-sinf (mol, diagramma)	Grafik tahlili	1
3	11-sinf 1-bob (yadro reaksiyalari)	Bilim va tushunish	1
4	8-sinf (Gey-Lyussak, Avogadro)	Hisoblash va masala yechish	1
5	8-sinf; Tirkashev *V* bob (oksidlanish darajasi)	Bilim va tushunish	1
6	8-sinf (atom tuzilishi)	Hisoblash va masala yechish	1
7	Tirkashev VI bob	Hisoblash va masala yechish	1
8	Tirkashev VI bob (Le-Shatelye)	Bilim va tushunish	1
9	7-sinf; Tirkashev (kristallogidrat)	Hisoblash va masala yechish	1
10	11-sinf 8-bob (Faradey qonunlari); Tirkashev *V* bob	Hisoblash va masala yechish	1

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11	8-sinf (atom tuzilishi, jadval)	Jadval tahlili	1
12	Tashev; 9-sinf (kuchlanish qatori)	Bilim va tushunish	1
13	8-sinf (indikatorlar)	Rasm bo‘yicha tahlil	1
14	9-sinf; Tashev (anorganik birikmalar xossalari)	Bilim va tushunish	1
15	9-sinf; Tirkashev (genetik zanjir/reaksiya turlari, hisob)	Hisoblash va masala yechish	1
16	Tashev (o‘g‘itlar)	Hisoblash va masala yechish	1
17	Tashev (nitratlar)	Bilim va tushunish	1
18	9-sinf; Tirkashev ($S/SO_2/H_2SO_4$, hisob)	Hisoblash va masala yechish	1
19	9-sinf; Tirkashev ($N/NH_3/HNO_3$, hisob)	Hisoblash va masala yechish	1
20	10-sinf §6, §12, §20, §28 (izomeriya)	Bilim va tushunish	1
21	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
22	10-sinf (aren, hisob)	Hisoblash va masala yechish	1
23	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
24	10-sinf §24; §26; §30 (kumush ko‘zgu)	Hisoblash va masala yechish	1
25	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
26	10-sinf (efir, hisob)	Hisoblash va masala yechish	1
27	10-sinf (organik birikmalar sinflari)	Bilim va tushunish	1
28	10-sinf §34	Hisoblash va masala yechish	1
29	10-sinf (pol, hisob)	Hisoblash va masala yechish	1
30	9-sinf (sifat reaksiyalar)	Bilim va tushunish	1
31	Tirkashev III bob (eritmalar)	Hisoblash va masala yechish	1
32	7-sinf (28-rasm); 8-sinf (stexiometriya)	Rasm bo‘yicha tahlil	1
33–35	Aralash masala (moslashtirish)	Ko‘p bosqichli hisob	3
36–40	Qisqa javobli masalalar	Hisoblash	5
41–43	Yozma ish	Usul va hisob (M, A)	75

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